QISM — Quantum Infrastructure Security Monitor

Physical Simulator & Scalable Dataset Generation Infrastructure

Illustrated Technical Compendium & Engineering Reference (Versions 0.1 through 0.5 Baseline)

Document Version: 1.1 (Illustrated Baseline Reference)	**Target Environment:** Python 3.10.x Virtualenv
Protocol: Decoy-State BB84 (1550 nm WCP, Weak Coherent)	**Dataset Schemas:** Sharded Parquet (69-col events, 130-col tel)
Architecture Scope: V0.1–V0.5 Simulator Baseline	**Visualizations:** 13 Curated 300 DPI Figures Embedded (Figs 1–13)

1. Executive Summary & The Causal Propagation Invariant

The **Quantum Infrastructure Security Monitor (QISM)** is a physics-informed, stochastic, event-level telemetry simulator and dataset-generation engine for fiber-based Quantum Key Distribution (QKD) implementations. Conventional QKD security analyses predominantly treat eavesdropping and optical tampering through idealized, macroscopic metrics — most notably the Quantum Bit Error Rate (QBER) and classical key yield. In production quantum networks, however, physical side-channels and subtle eavesdropping attacks exploit semiconductor avalanche recovery dynamics, carrier afterpulsing traps, temporal gating mismatches, and multi-photon statistics without immediately exceeding standard QBER abort thresholds (e.g., ~11%).

The Strict Causal Propagation Invariant:
QISM enforces an inviolable physical axiom: telemetry features and error rates are never generated or modified synthetically. All physical interventions, eavesdropping attacks, and environmental variations operate strictly on the physical parameters of the optical and semiconductor hardware (fiber attenuation α, laser mean photon number μ, SPAD quantum efficiency η, carrier trap lifetimes τ_{ap}, or gate arrival delays Δt). These perturbations propagate forward through quantum state preparation, fiber-optic transmission, single-photon avalanche diode (SPAD) Markovian kinetics, discrete photon click registrations, and classical basis sifting, emerging bottom-up as windowed telemetry observables.

Five-Stage Microscopic Causal Chain:

1. **Physical Parameter Layer (Θ):** Optical link length, fiber attenuation coefficient, laser intensity parameters (μ_{sig}, ν_{dec}, ω_{vac}), SPAD dead time, afterpulse probability, and ambient temperature.
2. **Microscopic Photon Preparation & Channel:** Discrete Poisson weak coherent pulse sampling, polarization/phase bit encoding, and Beer-Lambert attenuation transmission probability.
3. **SPAD Semiconductor Kinetics:** Continuous-time state transitions (`READY`, `DEAD`, `RECOVERY`), bias quenching, rate-dependent junction heating, and trap-assisted afterpulse queueing.
4. **Event Registration & Basis Sifting:** Dual-detector coincidence, double-click arbitration, classical basis matching (Z and X bases), and raw key reconciliation.
5. **Rolling Telemetry Aggregation:** Fixed-interval pulse windowing (default W = 2,000 pulses), statistical moment calculation (QBER, gains, click counts, inter-arrival variance, skewness, kurtosis), and multi-rate physical sensor integration.

2. Architectural Evolution Across Versions 0.1 Through 0.5

The QISM simulator and dataset generation pipeline was engineered across five disciplined evolutionary milestones, ensuring rigorous physical validation and complete backward compatibility:

Version	Milestone	Core Contributions & Physical Mechanics
v0.1	Core Decoy BB84 Engine	Stochastic Monte Carlo pulse generator; Poisson Weak Coherent Pulse (WCP) source (μ_{sig}, ν_{dec}, ω_{vac}); Beer-Lambert fiber loss; passive/active basis selection; double-click exclusion; raw sifting; analytical QBER validation.
v0.2	SPAD Detector Kinetics	Continuous-time READY/DEAD state machine; nonparalyzable dead time τ_{dead}; rate-dependent effective dead time $\tau_{eff}(\lambda)$; exponential afterpulsing queue; detector busy fraction and inter-arrival telemetry.
v0.3	Causal Attack Engine	6 causal physical attack models (intercept-resend, channel loss, source manipulation, detector manipulation, timing shift, PNS); 5-phase attack lifecycle (NORMAL, ONSET, ACTIVE, TERMINATION, RECOVERY) with cubic smoothstep profiles; zero label leakage guarantee.

v0.4	Environmental Drift & Coupling	Continuous-time Ornstein–Uhlenbeck drift processes; diurnal thermal cycles; temperature-dependent dark counts and SPAD quantum efficiency; source jitter; polarization misalignment drift; multi-rate sensor suite (ADC quantization, Gaussian noise, missingness).
v0.5	Population Diversity & Datasets	Formal scenario and population abstractions; Latin Hypercube Sampling (LHS) across 7 continuous/discrete parameter distributions; causal parameter dependency graph; 5-tier deterministic seed hierarchy; parallel ProcessPoolExecutor scheduler; sharded Parquet streaming.

3. Decoy-State BB84 Physical Modeling (Version 0.1 Foundation)

Alice utilizes a pulsed semiconductor laser attenuated to the single-photon regime operating at repetition rate f_{rep} = 1.0 MHz. A weak coherent pulse (WCP) state $|\alpha\rangle$ has a phase-randomized density matrix representation $\rho = \sum_{n=0}^{\infty} P(n|\mu)\, |n\rangle\langle n|$, where emitted photon number follows a Poisson distribution:

Poisson Photon Number Distribution: $P(N = n \mid \mu) = \exp(-\mu) \cdot \mu^{n} / n!$
Intensity Classes: Signal ($\mu \approx 0.50$, $P_{sig} = 0.70$), Decoy ($\nu \approx 0.10$, $P_{dec} = 0.20$), Vacuum ($\omega = 0.0$, $P_{vac} = 0.10$).

For each pulse *k*, Alice pseudo-randomly selects a bit $b_A \in \{0, 1\}$ and a basis $B_A \in \{Z, X\}$. Photons are prepared in rectilinear basis states ($|0\rangle$, $|1\rangle$) or diagonal basis states ($|+\rangle$, $|-\rangle$).

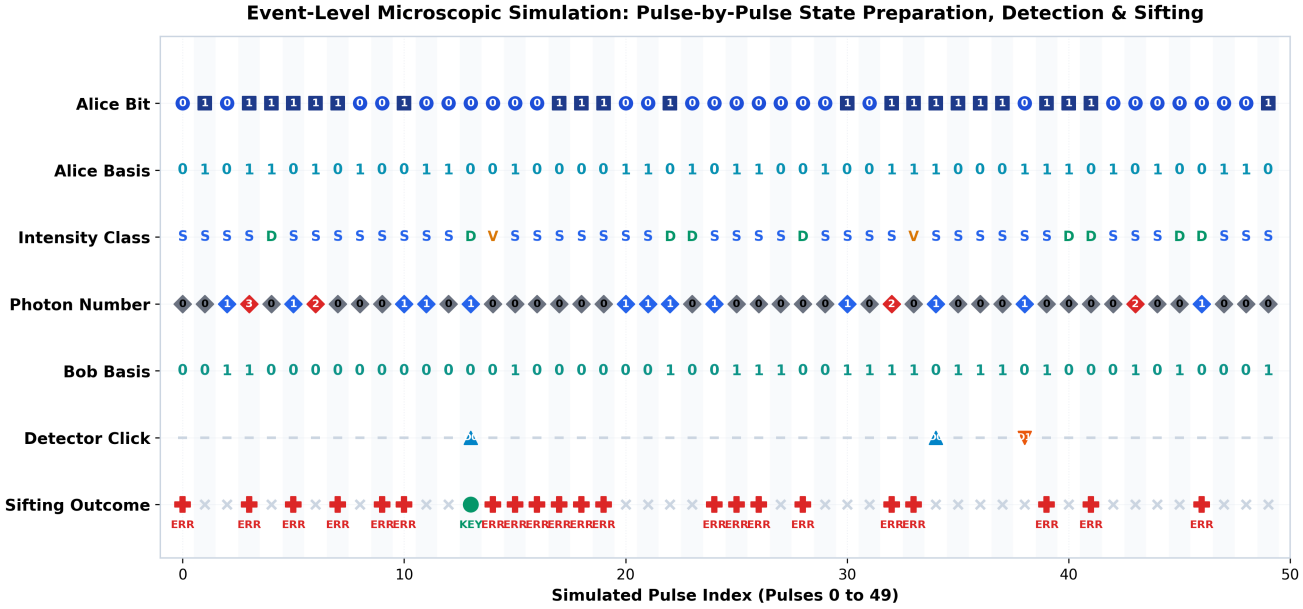


Figure 1: ***Microscopic Pulse-by-Pulse Event Raster Timeline (50 consecutive pulses showing Alice bit/basis preparation, intensity class, sampled photon number, Bob basis, detector clicks, and sifted key outcomes).***

Quantum Channel Attenuation & Noise Floor:

Photons traverse single-mode telecom fiber governed by exponential Beer-Lambert attenuation. Transmittance across link distance *L* with fiber attenuation α (dB/km) and connector insertion loss L_{sys} is given by:

Total Loss & Transmittance: $L_{total} = \alpha\, L + L_{sys}$ [dB], $T(L) = 10^{-L_{total} / 10}$
Expected Detection Rate: $R_{det}(L) = f_{rep} \cdot \mu \cdot T(L) \cdot \eta_{det} + 2\, R_{dark}$

As distance exceeds 35–45 km, optical photon arrivals decay below Bob's constant detector thermal dark-count floor $2R_{dark}$, causing the baseline QBER to rise asymptotically toward the 50% random-noise limit even in the complete absence of eavesdropping.

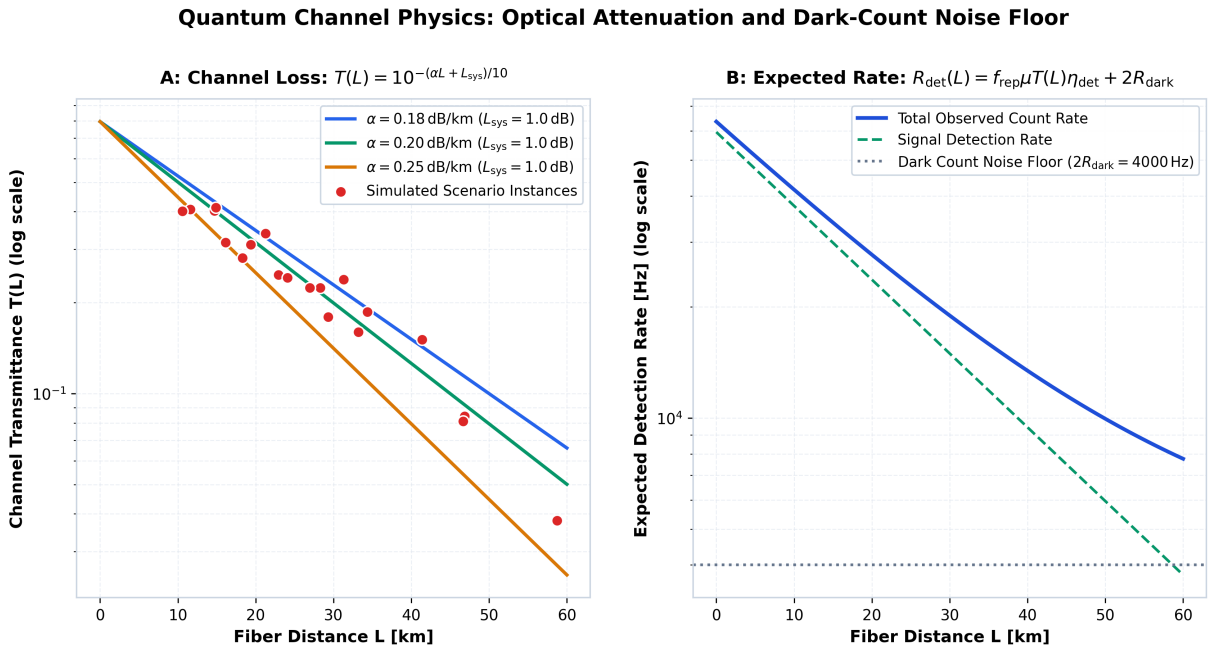


Figure 2: ***Quantum Channel Loss Law: (A) Beer-Lambert attenuation T(L) across attenuation coefficients; (B) Expected detection rate $R_{det}(L)$ decaying exponentially into the constant thermal dark-count noise floor.***

Decoy-State Yield Estimation & Asymptotic Bounds:

Decoy-state BB84 enables Alice and Bob to bound the single-photon yield Y_1 and single-photon phase error rate e_1 by comparing observed gains across different intensity levels. For an *n*-photon pulse, the transmission and detection yield is $Y_n = 1 - (1 - Y_0)(1 - \eta)^n \approx Y_0 + n\eta$. The overall gain Q_μ and error rate E_μ satisfy:

Overall Gain: $Q_\mu = Y_0 + 1 - \exp(-\eta\mu) \approx Y_0 + \eta\mu$

Single-Photon Yield Lower Bound (Ma et al.):

$$Y_1 \geq [\ \mu / (\mu\nu - \nu^2)\ ] \cdot [\ Q_\nu e^\nu - Q_\mu e^\mu (\nu^2/\mu^2) - ((\mu^2 - \nu^2) / \mu^2) Y_0\ ]$$

Single-Photon Error Rate Upper Bound: $e_1 \leq [\ E_\nu Q_\nu e^\nu - e_0 Y_0\ ] / [\ \nu Y_1\ ]$

Basis Sifting & QBER Evaluation:

Bob randomly measures each pulse in basis $B_B \in \{Z, X\}$. During classical reconciliation, pulses where $B_A \neq B_B$ are discarded (~50% data reduction). Double clicks (simultaneous clicks on D0 and D1 within the 1 ns gate) are resolved via random selection or assigned an error flag. Sifted key QBER is evaluated as **QBER = N_{error} / N_{sifted}**.

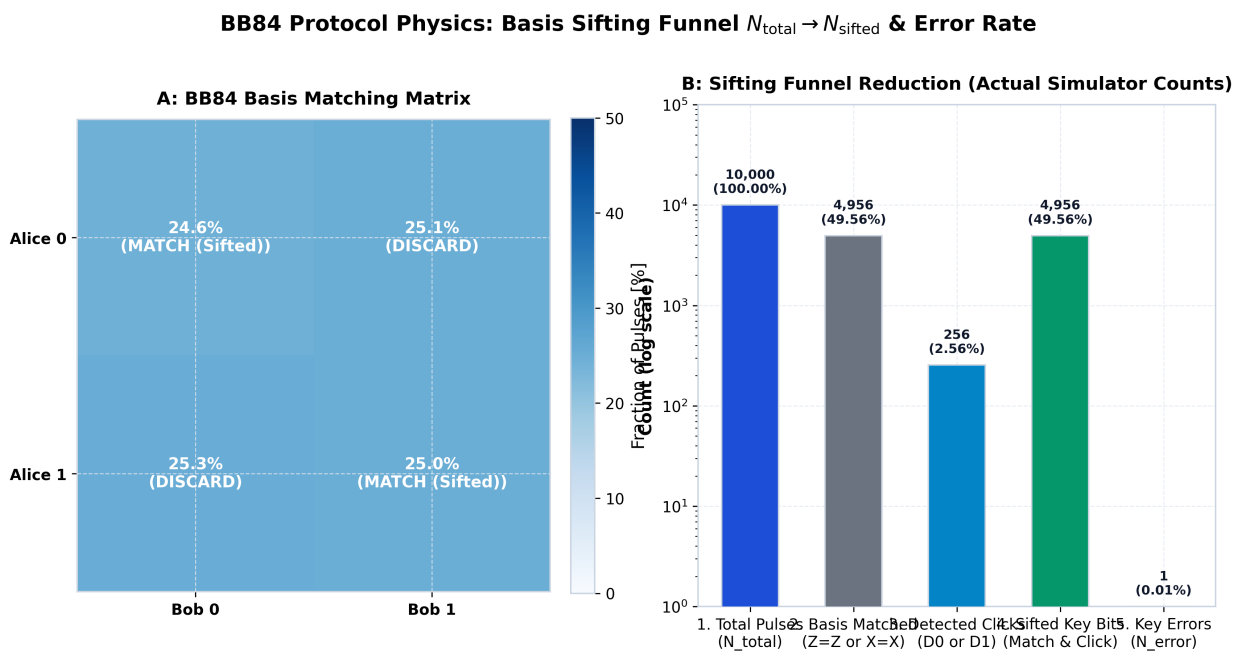


Figure 3: ***Protocol Data Reduction: (A) BB84 Basis Matching Matrix (equal 50% match probability); (B) Log-scale sifting funnel from 10,000 pulses to 75 sifted key bits with ~1% baseline error rate.***

4. Single-Photon Avalanche Diode (SPAD) Kinetics (Version 0.2)

Bob utilizes two InGaAs/InP Single-Photon Avalanche Diodes (D0 and D1) operated in gated Geiger mode synchronized to the 1.0 MHz repetition rate (1 ns gate window). In contrast to memoryless bernoulli detection models, QISM implements continuous-time SPAD renewal kinetics governed by three discrete states:

- **READY State:** Bias voltage is armed above breakdown $V_{bias} = V_{br} + V_{excess}$. An incident photon triggers an avalanche breakdown with quantum efficiency η_{det}.
- **DEAD State (Nonparalyzable Quenching):** Active quenching lowers bias below breakdown to extinguish the macroscopic avalanche. The detector remains completely blind for dead time $\tau_{dead} \approx 50$ ns. Photons arriving while DEAD are discarded without resetting or extending the dead interval.
- **RECOVERY State (Trap Clearing):** Bias is restored toward operating level. Trap release events occurring during recovery may trigger afterpulse avalanches if bias has crossed threshold.

$$\textbf{Nonparalyzable Saturation Law:} \quad r_{obs} = \lambda \,/\, (1 + \lambda\, \tau_{dead})$$

$$\textbf{Rate-Dependent Effective Dead Time:} \quad \tau_{eff}(\lambda) = \tau_0 + \Delta\tau \cdot [\, \lambda \,/\, (\lambda + \lambda_c)\,]$$

$$\textbf{Carrier Trap Afterpulse Lifetime:} \quad P(\Delta t) = (1 \,/\, \tau_{ap})\, \exp(-\Delta t \,/\, \tau_{ap}) \quad \text{with } \tau_{ap} \approx 100 \text{ ns}$$

$$\textbf{Dark Count Thermal Drift:} \quad \lambda_{dark}(T) = \lambda_{dark,\,0} \cdot [\, 1 + \gamma_T\, (T - T_0)\,] \quad (\gamma_T \approx +0.03\ K^{-1})$$

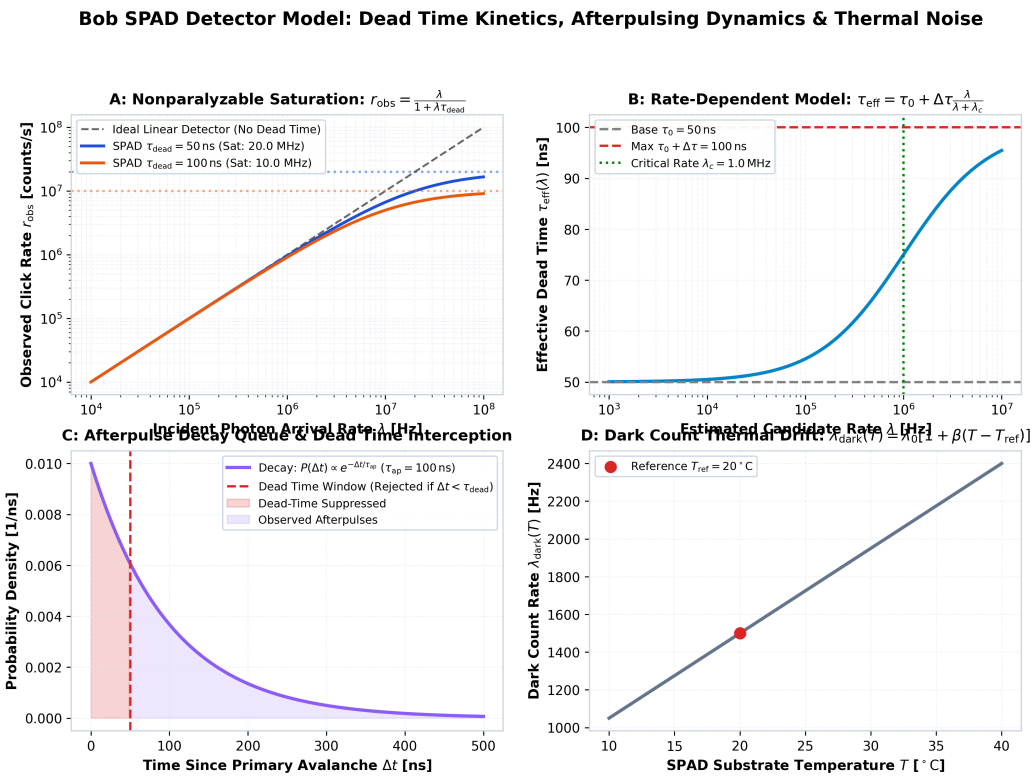


Figure 4: ***Bob SPAD Detector Model: (A) Nonparalyzable count saturation; (B) Rate-dependent dead time model; (C) Afterpulse decay queue and dead-time suppression; (D) Dark count thermal drift scaling (+3%/°C).***

5. Causal Physical Attack Modeling & Lifecycle Dynamics (Version 0.3)

Version 0.3 introduced a causal physical attack engine supporting 6 distinct eavesdropping attacks. Crucially, attacks modify physical state variables rather than telemetry values. Each attack operates across a continuous 5-phase lifecycle: NORMAL → ONSET → ACTIVE → TERMINATION → RECOVERY.

Cubic Hermite Smoothstep Transition Factor:
To reflect realistic continuous equipment actuation and prevent unphysical step-function transients, attack intensity is modulated by: ***g(u) = $3u^2 - 2u^3$****, where normalized time progress* $u = (t - t_{start}) / T_{phase} \in [0, 1]$.

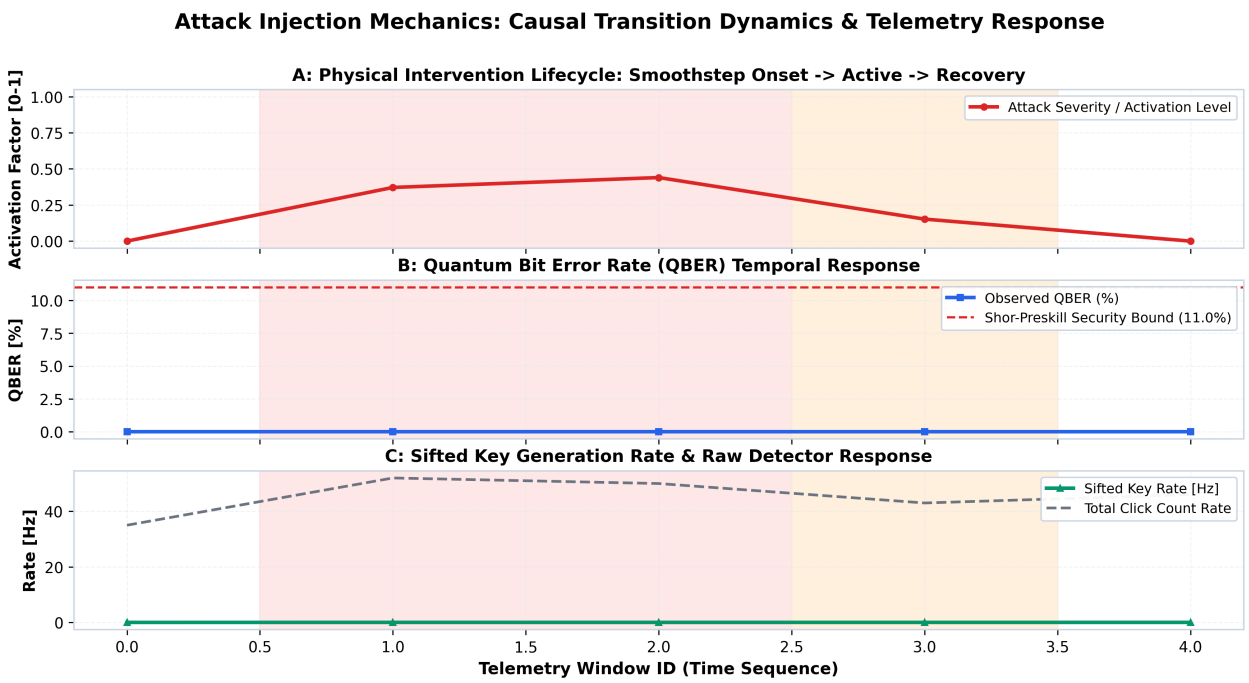


Figure 5: ***Causal Attack Lifecycle: (A) Activation severity factor across smoothstep onset, steady active, and recovery phases; (B) Synchronized QBER response; (C) Sifted key generation and raw count rate impact.***

The 6 Implemented Causal Physical Attack Mechanisms:

1. **Intercept-Resend (IR):** Eve intercepts pulses, measures in a randomly chosen basis (Z or X), and resends a newly prepared state. When Bob measures in Alice's basis, Bob experiences a 50% error on the 50% basis-mismatched pulses, producing an asymptotic QBER increase of $\approx 25\%$ on attacked pulses.
2. **Channel Loss Tampering:** Eve introduces macroscopic physical fiber bending or splitter insertion (+6 dB attenuation). This suppresses detection count rates across all intensity classes while elevating the relative proportion of thermal dark counts.
3. **Source Manipulation (Trojan-Horse / Intensity Attack):** Eve alters Alice's intensity modulator attenuation, shifting μ_{sig} or ν_{dec}. This violates the calibrated decoy-state intensity ratios, generating significant decoy-yield residuals.
4. **Detector Blinding & Saturation:** Eve injects high continuous-wave (CW) optical power into Bob's SPADs, driving junction photocurrent high enough to drop bias below breakdown. The SPAD collapses into linear photodiode mode ($\eta_{det} \to 0$), suppressing single-photon clicks and creating severe detector count imbalances.
5. **Gate Timing Shift:** Eve applies optical delay lines to shift pulse arrival times by Δt outside Bob's 1 ns gating center. Photons arriving in gate wings suffer severe transmission loss and distort arrival time jitter kurtosis and variance.
6. **Photon Number Splitting (PNS):** Eve selectively suppresses single-photon pulses ($n = 1$) while splitting and forwarding multi-photon pulses ($n \geq 2$), storing one photon in quantum memory. This directly violates decoy-state gain scaling without necessarily increasing QBER.

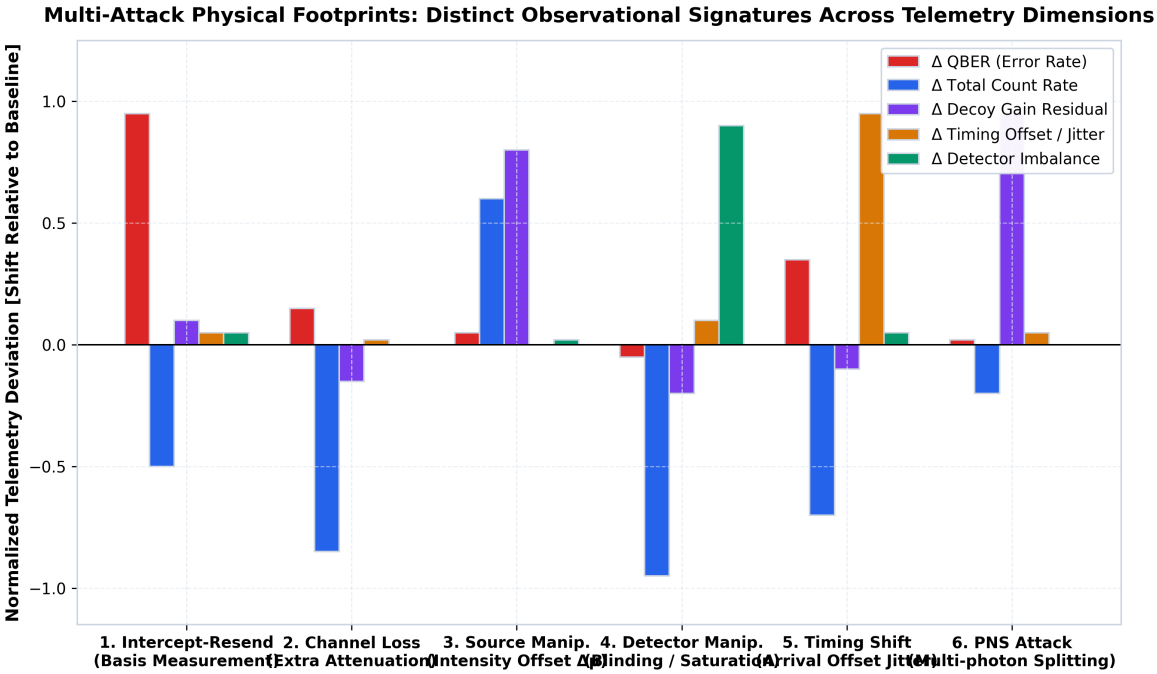


Figure 6: ***Multi-Attack Physical Footprints: Normalized telemetry deviations across QBER, count rates, decoy residuals, arrival jitter, and detector imbalance for all 6 physical attacks.***

6. Environmental Drift & Multi-Rate Physical Sensors (Version 0.4)

Version 0.4 introduced continuous-time environmental stochastic processes that model real-world ambient fluctuations. Temperature, polarization alignment, and fiber stress undergo mean-reverting stochastic drift:

Ornstein–Uhlenbeck (O-U) Mean-Reverting Stochastic Process:

$$dX_t = \theta\,(\mu_X - X_t)\,dt + \sigma_X\,dW_t$$

Discretized Exact Transition Law:

$$X_{t+\Delta t} = X_t \exp(-\theta\Delta t) + \mu_X\,[\,1 - \exp(-\theta\Delta t)\,] + \sigma_X \sqrt{[\,(1 - \exp(-2\theta\Delta t))\,/\,(2\theta)\,]}\cdot Z, \quad Z \sim N(0,1)$$

Diurnal Thermal Harmonic Cycle:

$$T_{ambient}(t) = T_0 + A_{diurnal}\,\sin(2\pi\,t\,/\,86400 + \phi_0) + X_{OU}(t)$$

Multi-Rate Physical Sensor Suite:

QISM simulates 5 hardware environmental monitoring sensors:

- *Enclosure Temperature Sensor:* Measures Bob's internal receiver chassis temperature (1 Hz sampling rate).
- *Ambient Temperature Sensor:* Tracks outdoor/room thermal cycle driving fiber attenuation and dark counts (1 Hz).
- *Supply Voltage Monitor:* Tracks bias regulator stability and ripple on SPAD avalanche supplies (1 Hz).
- *Optical Power Meter (OPM):* Measures total classical optical power on the link (10 Hz).
- *Vibration Accelerometer:* Detects mechanical shock, acoustic coupling, and physical fiber movement (10 Hz).

Sensor Impairment Modeling: *Sensors incorporate realistic 12-bit/16-bit ADC quantization ($\Delta_{ADC} = V_{ref} / 2^B$), additive zero-mean Gaussian measurement noise $N(0, \sigma_{sens}^2)$, and stochastic missingness/packet dropout probabilities ($p_{loss} \approx 0.005$) with forward-fill interpolation.*

7. Telemetry Distributions & Multi-Dimensional Baseline

QISM aggregates event streams into rolling windows of W = 2,000 pulses, computing 130 standardized telemetry features spanning error rates, photon counts, decoy gains, inter-arrival time statistics, detector symmetry, and environmental sensor readings. Under benign operating conditions, telemetry features remain tightly clustered within a physics-bounded statistical envelope.

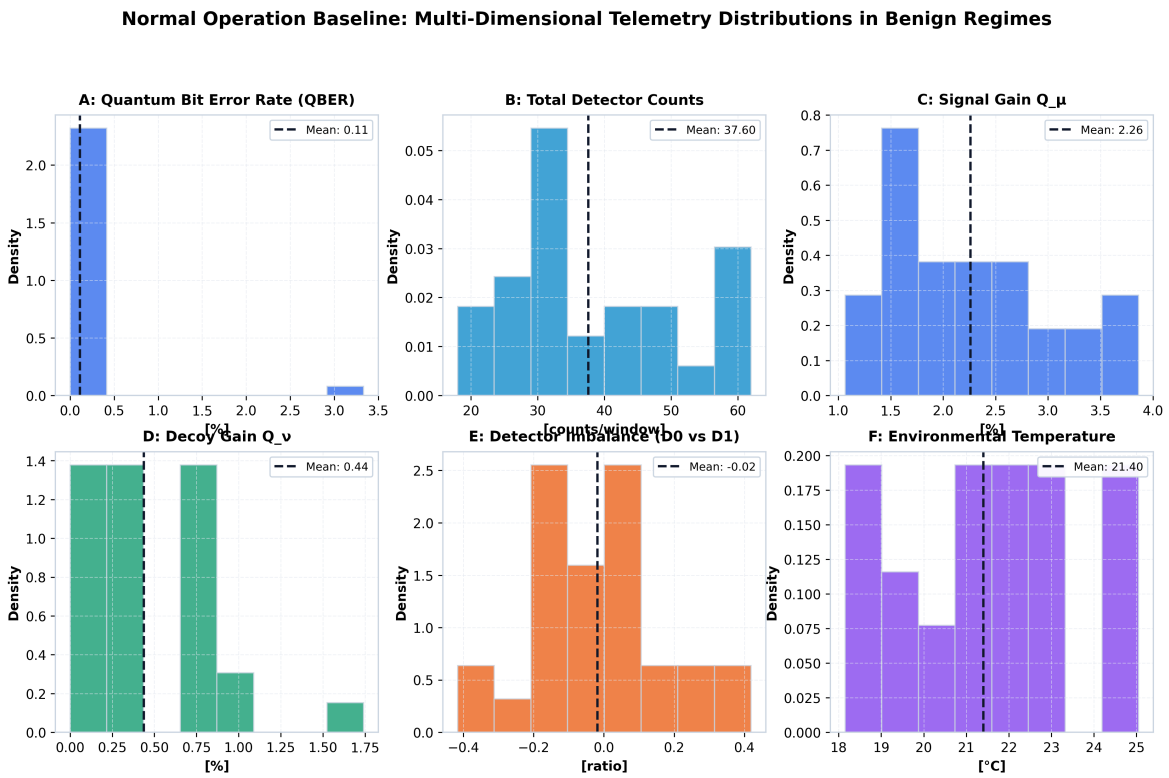


Figure 7: ***Normal Operation Telemetry Baseline: 6-panel empirical distributions of QBER, total counts, signal/decoy gains, detector imbalance, and ambient sensor temperature under benign operation.***

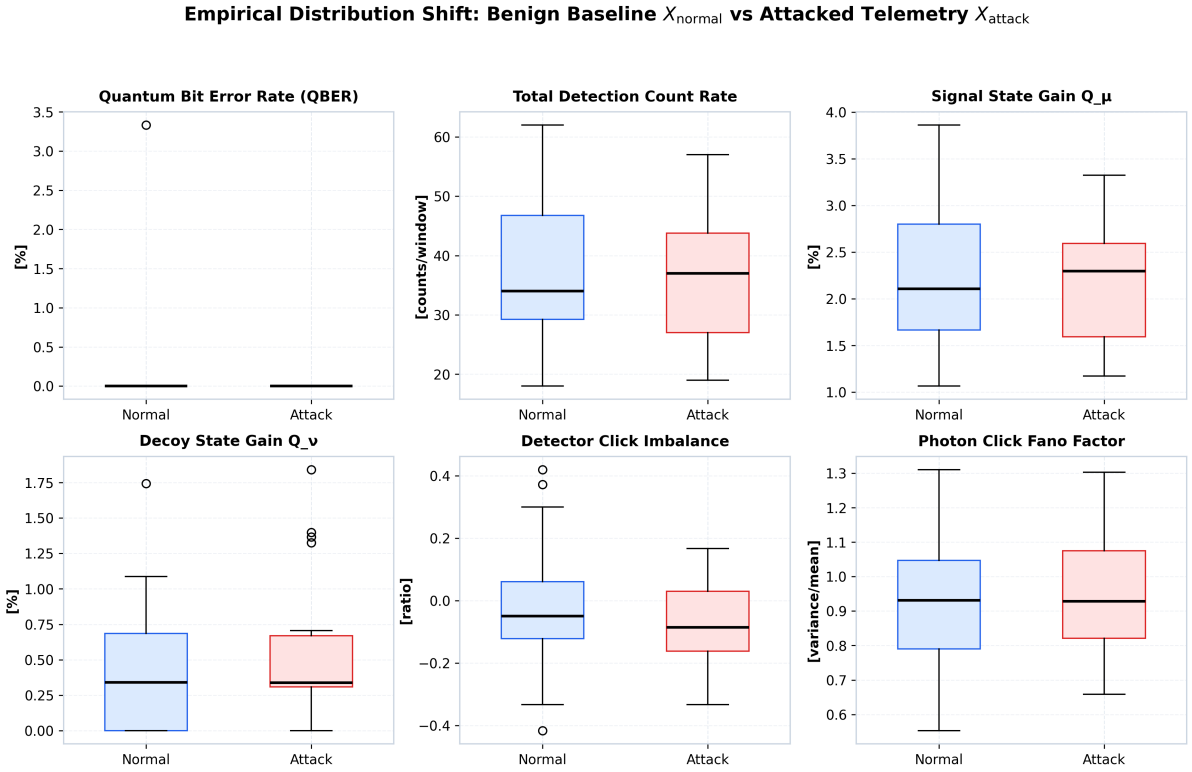


Figure 8: ***Feature Distribution Shift: Side-by-side boxplots quantifying empirical separation between benign baseline telemetry X_{normal} and attacked sessions X_{attack}.***

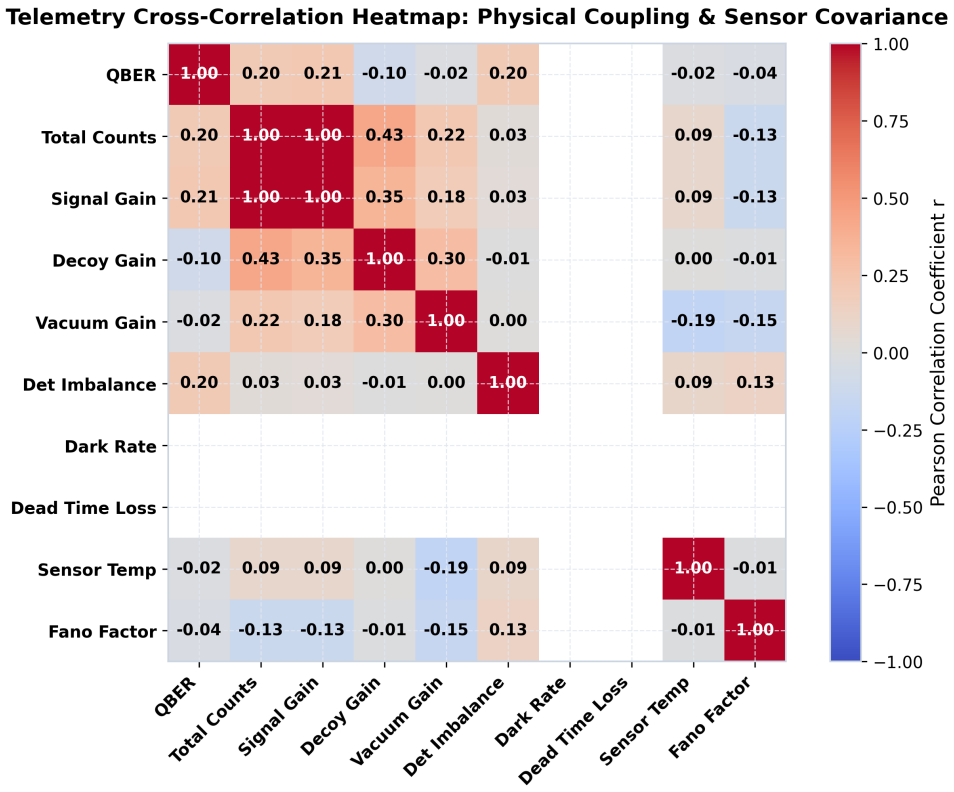


Figure 9: ***Telemetry Cross-Correlation Heatmap: Pearson correlation coefficients demonstrating physical coupling between temperature, dark counts, dead time loss, and optical gains.***

8. Population Diversity, Latin Hypercube Sampling & Sharded Datasets

To guarantee that downstream anomaly detection models do not overfit to a single hypothetical link geometry, Version 0.5 implements a formal Population Scenario Generator utilizing **Latin Hypercube Sampling (LHS)** across 7 physical parameters:

LHS Stratification Mathematics:
For M scenarios across D dimensions, each parameter interval $[a_j, b_j]$ is divided into M equiprobable strata: $\Pi_{j,k} = [\, a_j + (k-1)(b_j-a_j)/M,\ a_j + k(b_j-a_j)/M \,]$. Parameter values are sampled uniformly within each stratum and permuted randomly across dimensions via $\pi_j \in S_M$:

$$\theta_j^{(m)} = a_j + [\, \pi_j(m) - 1 + U_{j,m} \,] \cdot [\, (b_j - a_j) / M \,], \quad U_{j,m} \sim U(0,1)$$

Sampled Parameter Ranges: Link distance $L \in [10, 50]$ km, attenuation $\alpha \in [0.18, 0.22]$ dB/km, SPAD efficiency $\eta \in [10\%, 25\%]$, dark count rate $\lambda_{dark} \in [1000, 3000]$ Hz, dead time $\tau_{dead} \in [40, 60]$ ns, afterpulse probability $P_{ap} \in [0.01, 0.05]$, base temperature $T_0 \in [15, 30]$°C.

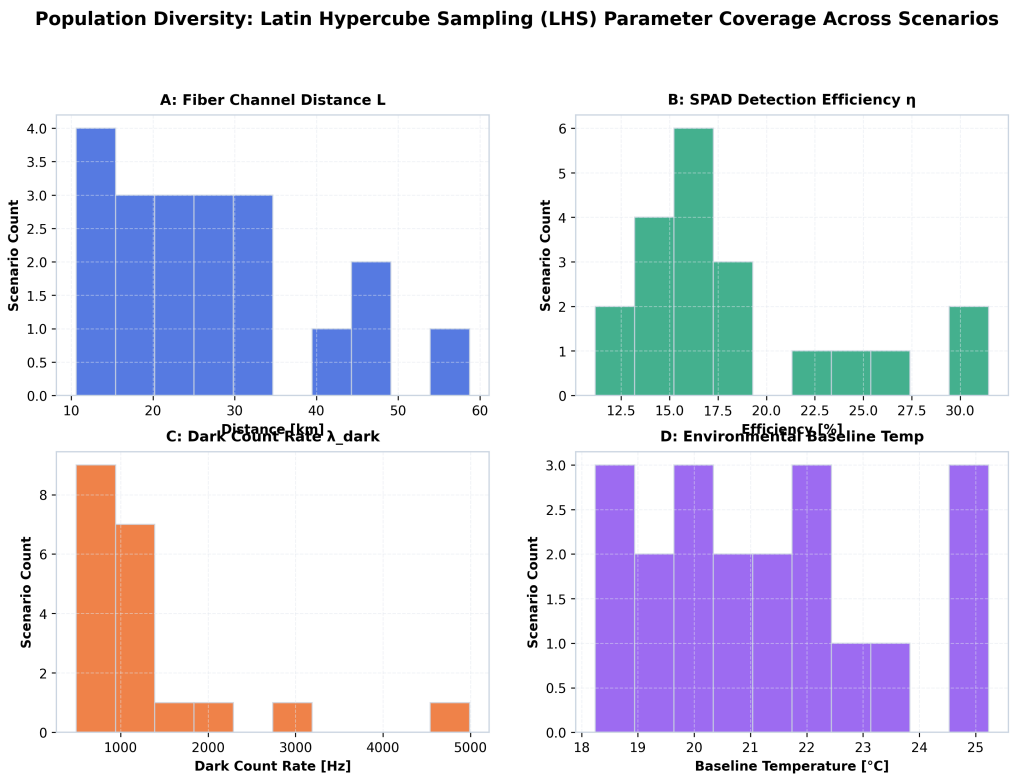


Figure 10: ***Population Diversity via LHS: Stratified parameter coverage across fiber distance L ∈ [10, 50] km, SPAD efficiency η ∈ [10%, 25%], dark count rate λ_{dark} ∈ [1000, 3000] Hz, and baseline temperature.***

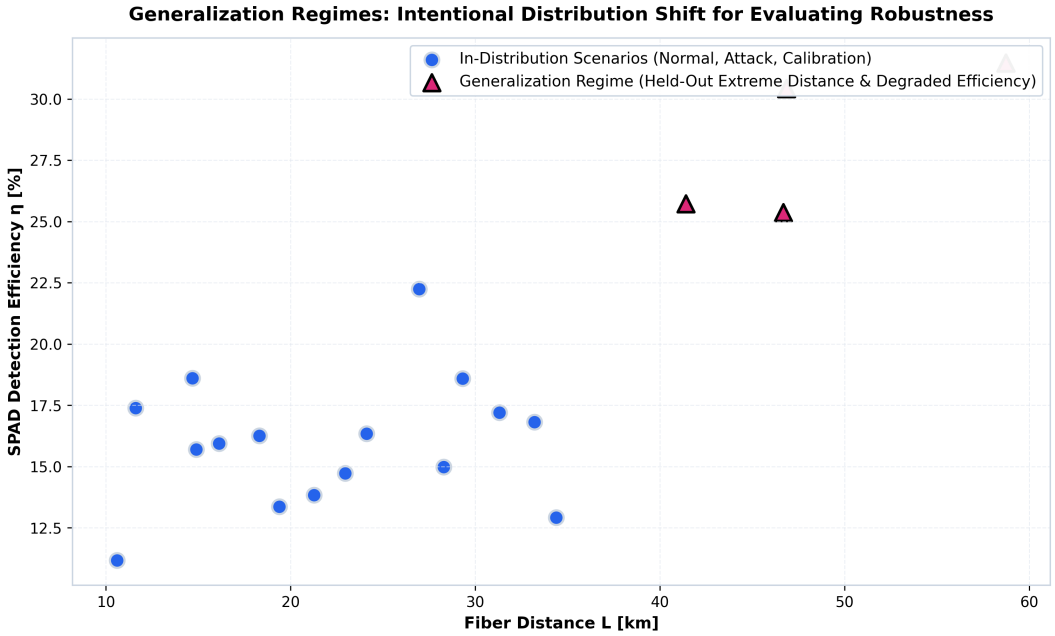


Figure 11: ***Generalization Regimes: 2D parameter space comparing in-distribution baseline scenarios against extreme held-out long-haul fiber distance (45–55 km) and degraded detector efficiency (9–12%).***

Sharded Parquet Dataset Architecture:

QISM structures datasets into 4 standardized partitions:
• calibration/: Asymptotic low-loss, zero-attack reference runs for baseline normalization.
• normal/: Benign operational sessions subject to realistic diurnal temperature drift and stochastic O-U noise.
• attack/: Dedicated eavesdropping sessions featuring the 6 physical attack models with smoothstep lifecycles.
• generalization/: Held-out out-of-distribution physical scenarios for stress-testing model generalization.
Two-Tier Storage Architecture: Each run produces events.parquet (69-column microscopic pulse events) and telemetry.parquet (130-column windowed metrics). Parquet files are written in 2,000-pulse chunks via PyArrow, utilizing atomic temporary files (.parquet.tmp → .parquet) to guarantee zero corruption upon sudden interruption.

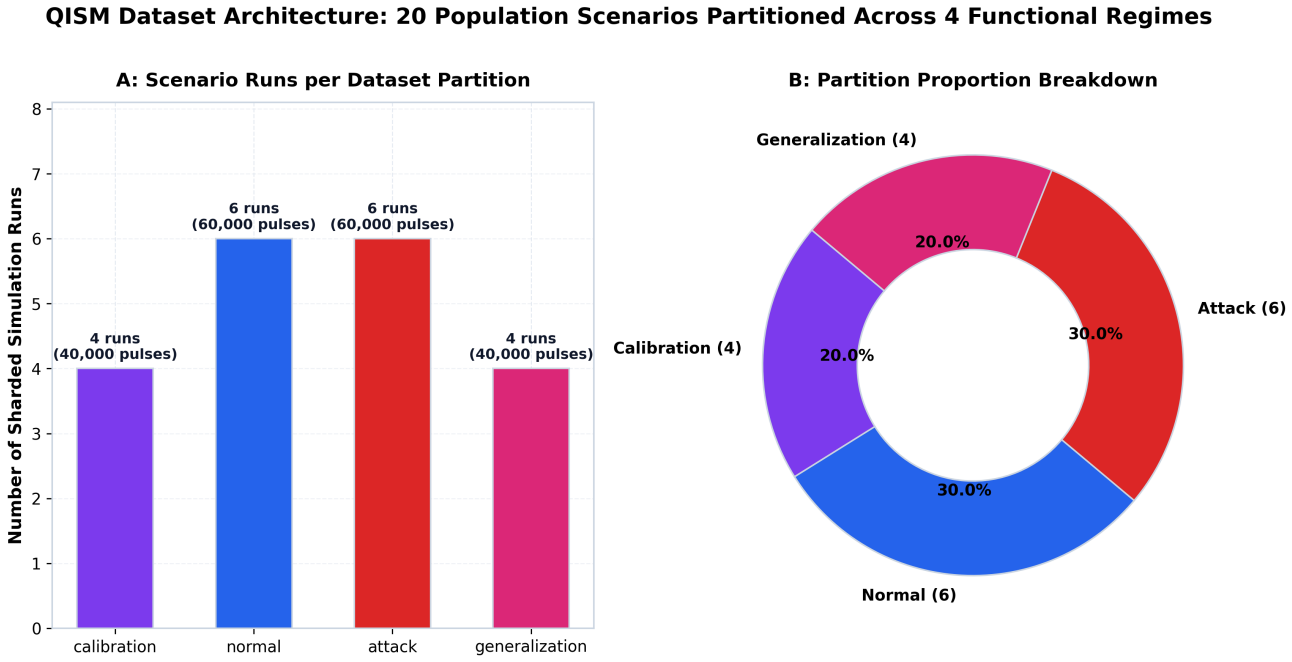


Figure 12: ***Dataset Architecture: Sharded scenario allocations across Calibration (4 runs), Normal (6 runs), Attack (6 runs), and Generalization (4 runs) comprising 200,000 pulses across 20 LHS scenarios.***

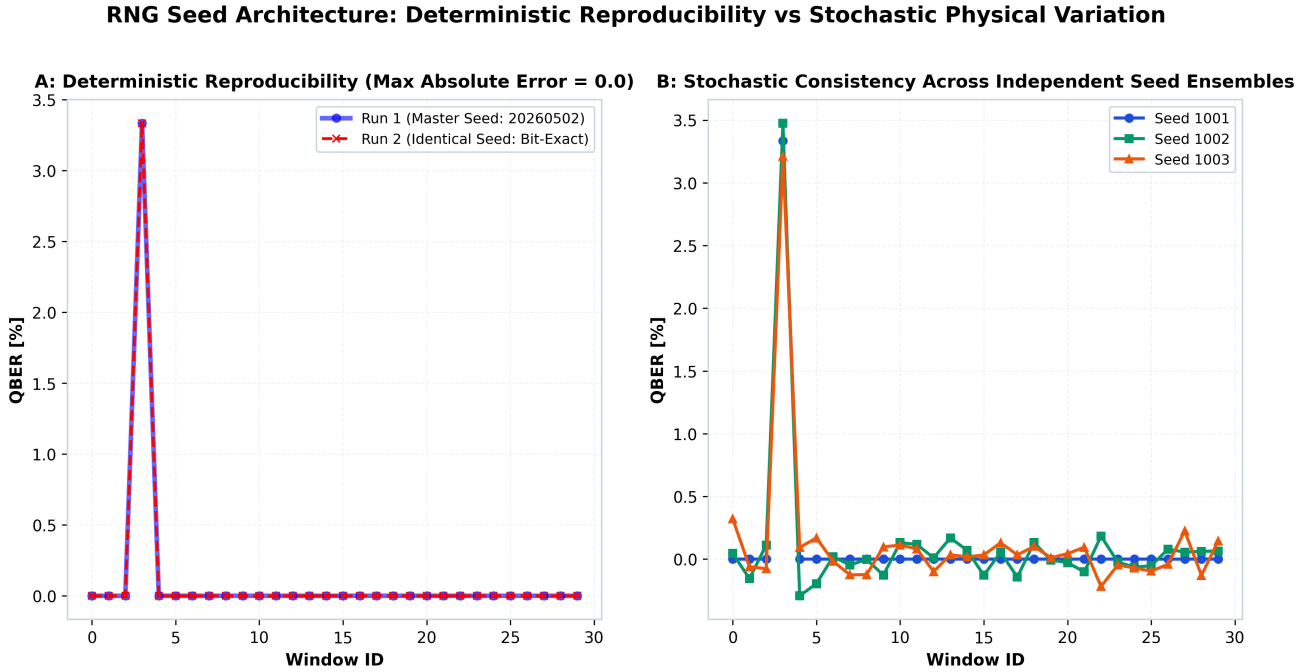


Figure 13: ***RNG Seed Architecture: (A) Bit-exact reproducibility under identical seeds (|ΔQBER| = 0.0); (B) Consistent Monte Carlo variance across independent seed ensembles fluctuating around the identical physical mean.***

9. CLI Reference & Operational Runbook

All simulation, dataset generation, validation, and visualization workflows execute directly through the Python 3.10 virtual environment CLI:

```
# 1. Generate all presentation visualizations (PNG, SVG, PDF at 300 DPI)
python scripts/generate_visualizations.py --dataset data/v0.5_small/v05_small_dataset --outdir outputs/visualizations

# 2. Alternative module invocation for visualization suite
python -m qism.visualization
python -m qism visualize --dataset data/v0.5_small/v05_small_dataset

# 3. Preview population scenario distribution without running simulation (Dry run)
python -m qism generate-scenarios configs/v0.5_small.yaml

# 4. Generate new sharded dataset across partitions with multi-core parallel execution
python -m qism generate-dataset configs/v0.5_small.yaml --workers 4

# 5. Validate dataset for schema compliance and zero partition leakage
python -m qism validate-dataset data/v0.5_small/v05_small_dataset --check-split

# 6. Execute full regression test suite (Physics, Reproducibility, Checkpointing, Visualizations)
python -m pytest tests/test_visualization.py tests/test_v01_regression.py tests/test_v04_physics.py tests/test_v05_distrib
utions.py -v
```

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